September 29

Last time: we talked about linear regression as a tool for exploring the relationship between 2 variables

We had some variable X, which we called the predictor, and another var Y which we called the response

We assume there is some true linear relationship between those 2 variables in the overall population

assume: population relationship is linear

We don't know the true population relationship, but we want to estimate it from data

* don't know true line
* estimate from data $(x_1, y_1), \ldots, (x_n, y_n)$

The data we observe consist of pairs of observed values of x and y. From the observed data, we can estimate the true population relationship btwn x and y.

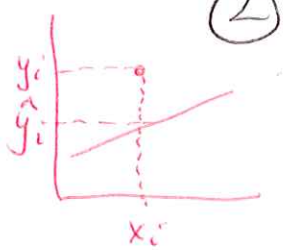
How to estimate?

We started out by visualizing the data points on a scatter plot, and said we wanted to draw a line through the scatter plot that comes closest to the data pts.

find line "closest" to data

Our definition of "closest" was the line that minimized the sum of squared vertical distances from the data points to the ~~kk~~ line.

"closest": minimizes $\sum_{i=1}^{n} (y_i - \hat{y}_i)^2$ vertical distances



We called the estimated slope $\hat{\beta}_1$ and the estimated intercept $\hat{\beta}_0$, so the estimated regression line is

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$$

($\hat{\beta}_0$: intercept; $\hat{\beta}_1$: slope)

The point on the regression line for a particular value of x is $(x_i, \hat{y}_i)$ point on line: $(x_i, \hat{y}_i)$

In the true population model, the data points are scattered around the line due to random error terms:

population model: $y_i = \beta_0 + \beta_1 x + \underbrace{\varepsilon}_{\text{random error}}$

The error terms are random, and their expected value is 0.

$$\mathbb{E}(\varepsilon_{\text{random}}) = 0$$

That means that for a particular value of x, the expected value of y is $\beta_0 + \beta_1 x$

$$\mathbb{E}(Y \mid X = x_0) = \beta_0 + \beta_1 x$$

So our prediction $\hat{y}_i$ corresponds to our estimate of the "average" or "typical" value of the response variable for that value of x.

One example of a linear relationship we talked about was SAT scores and GPA. There is some true relationship in the

Overall population, which we don't know.

population: $GPA = \beta_0 + \beta_1 * SAT + \varepsilon$

We collect a random sample of pairs of SAT scores & GPAs, and ~~observe~~ fit a regression line to the ~~observed~~ sample data.

observe $(x_1, y_1), \ldots, (x_n, y_n)$

estimate: $\hat{\beta}_0 = 0.6$, $\hat{\beta}_1 = 0.0025$

We get an estimated regression line with intercept 0.6 and slope 0.0025. What does that mean? What is the interpretation of the fitted model?

interpretation?

plug in $X = 1000$.

$0.6 + 0.0025 * 1000 = 3.1$

If we plug in an SAT score of 1000, we get a prediction of... 3.1 for the GPA.

According to the model, on average we expect people who get SAT scores of 1000 to have first-year GPAs of 3.1.

If SAT = 1000, "typical" GPA is 3.1

What about the intercept? Technically the intercept represents our guess about the typical value of y when $x=0$. So, we're estimating that on average, students who scored 0 on the SAT have GPAs of 0.6

$\hat{\beta}_0 = 0.6$ means 0.6 is "typical" GPA when SAT = 0

X not meaningful (min possible SAT is 400)

This isn't meaningful, because the min possible SAT score is 400. Depending on context, the intercept may or may not have a useful interpretation.

slope

$\hat{\beta}_1 = 0.0025$

What about the slope? The slope of the fitted line is 0.0025. For students whose SAT scores differ by 100 points, on average we expect their ~~GPA~~ GPAs to differ by 0.25 pts.

students w/SAT scores that differ by 100 pts ~~have~~ have GPAs that differ by 0.25, on average.

There are many ways we could phrase this.
"For students whose SAT scores differ by 100 points, we predict their GPAs differ by 0.25 points."

(Note - no special reason we're talking about a 100pt SAT difference, just makes the scale of the numbers convenient).

DON'T imply causal relationship.

What we don't want to do is phrase it in a way that implies a CAUSAL relationship btwn SAT score & GPA.

eg "increasing SAT score by 100 pts tends to increase GPA by 0.25 pts"; or "a 100 pt increase in SAT score is associated with an 0.25pt increase in GPA."

don't say or "0.0025 is the increase in GPA due to 1pt increase in SAT score"

You've likely heard before that "correlation does not imply causation"

Correlation ≠ causation

In this example, that is already fairly obvious: we don't think SAT scores cause GPA. There is an association because they both try to measure something about academic performance.

SAT score doesn't cause GPA (not always this obvious)

Sometimes it won't be this obvious, and it might be tempting to conclude that there is a causal relationship between X and Y.

Let's consider an example from developmental psychology.

ex: language acquisition

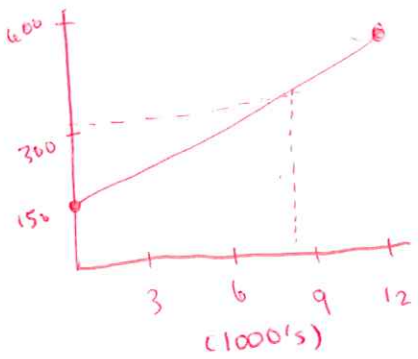
Researchers are interested in how babies develop language skills. We want to consider the relationship between the # words spoken to an infant each day, and their vocabulary size @ age 24 months.

X = # infant-directed words

Y = vocab size @ 24mo

Suppose we go observe 30 households, and collect audio recordings. Some parents talk to their babies more, some less. The values ranged from 1,000 to 12,000.

observe $(X_1, Y_1), \ldots, (X_{30}, Y_{30})$ $\quad 1k \leq X_i \leq 12k$



We fit a regression line to the sample data, and get an estimated slope of 17.5 and est. intercept of 150

$$\hat{y} = 150 + 17.5x$$

How would we interpret this model?

interpretation?

For infants who have 8000 words spoken to them daily, a typical vocabulary at 24mo would be $150 + 17.5 * 8 = 310$

hear 8000 words → "typical" vocab size $= 150 + 17.5 * 8$
$= 310$

What about the intercept? Technically, the intercept of 150 says that we expect infants who have ZERO words spoken to them daily, to have a vocabulary of 150 words @ age 24mo.

intercept: expected vocab size for infants hearing 0 words
$= 150$

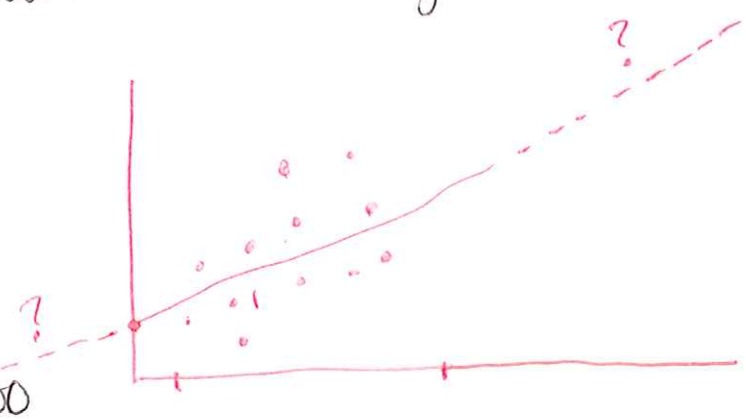
Let's talk about **extrapolation** for a minute.
We're estimating the relationship between X and Y with a line. Mathematically, ~~[illegible]~~ we COULD plug in any x value and get a predicted y value for y.

could plug in any x value, get $\hat{y}$

But in practice, it can be a bad idea to use the regression line to make predictions for outside the range of x values we actually observed.

doesn't mean we should

In the SAT example that was obvious, because SAT scores are constrained to be between 400 & 1600



In the language acquisition study, the observed households had x values ranging from 1k words/day to 12k words/day. For 0 words/day or 50k words/day, those scenarios are very different from the ones we observed, so it might be unreasonable to think our model can also make good predictions for those values. Do we really think the linear relationship extends for ALL values of x?

observed x between 1000 & 12000
↳ linear relationship might not extend for ALL x
↳ using our model to predict for x=0 or x=50,000 is risky

Making predictions outside the range of the observed x values is called "extrapolation", and can lead to flawed conclusions.

"extrapolation" - making predictions @ x values outside observed range

↳ often unwise

What about the interpretation of the estimated slope?

$\hat{\beta}_1 = 17.5$ interpretation?

Unlike SAT/GPA example, in this case it does seem intuitively plausible that the # words spoken to an infant directly affects their vocabulary size.

infant-directed words MIGHT affect vocab

It might be tempting to say "We estimate that speaking an extra 1000 words to infant increases vocabulary size by 17.5 words."

don't say:
" We estimate speaking 1000 extra words increases vocab by 17.5 "

But describing the relationship that way implies that speaking the 1000 extra words CAUSES infant's vocabulary size to grow by 17.5 words.

→ implies causal relationship

We would be justified in saying "on avg, babies know 17.5 more words at age 2, compared to other babies who have 1000 fewer words spoken to them daily"; or

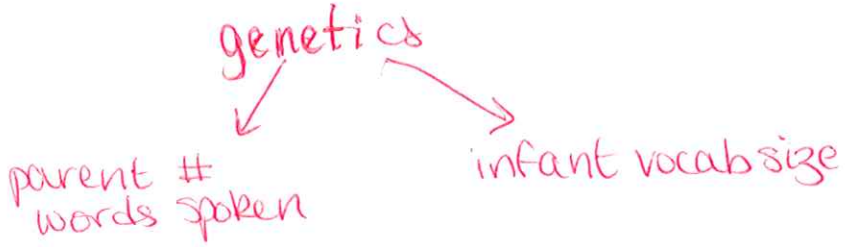
ok: "babies whose infant-dir. speech differs by 1000 words tend to have vocab size that differs by 17.5 words."

We can't draw conclusions about causality, because the data were collected in an observational study. In an observational study, we can't rule out other possible reasons for the observed association between X and Y.

observational studies – don't show causality

eg maybe language skills are mostly genetic. Parents w/weaker language skills choose not to speak to their babies as much, and due to the genetic trait, their babies also have smaller vocabularies. Then it wouldn't be true that having the parent speak an extra 1000 words/day to the baby would increase their vocab by 17.5 words, because their vocab size is being determined by genetics.

maybe:



If we wanted to assess whether # **infant**-directed words causes a difference in vocab size, would need to design an experiment observing a group of families whose ONLY difference was # infant-directed words, otherwise have all the same **characteristics**. Then we could conclude the difference in vocab size was BECAUSE of the difference in # infant directed words.

to make causal conclusions: need a controlled experiment

Overall, that is how we interpret the fitted model (the coefficients, the points on the line) in context.

In a purely mathematical way, we can also say something interesting about how the slope estimate is calculated.

$\hat{\beta}_1$ mathematical interpretation

If we rearrange the formula for $\hat{\beta}_1$, we can show that...

$$\hat{\beta}_1 = \frac{\sum_{i=1}^{n}(x_i-\bar{x})(y_i-\bar{y})}{\sum_{i=1}^{n}(x_i-\bar{x})^2} = \text{sample correlation}(x,y) * \frac{\text{sample stdev}(y)}{\text{sample stdev}(x)}$$

Sample correlation is related to our earlier discussion about covariance/correlation of random variables $E((x-\bar{x})(y-\bar{y}))$. There is a counterpart that we calculate for sample data.

$$r_{xy} = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2}\sqrt{\sum_{i=1}^{n}(y_i - \bar{y})^2}}$$ sample correlation of X & Y

If X and Y are very strongly correlated, and the correlation is positive, then X and Y tend to take large values at the same time.

strong, pos correlation: X & Y big @ same time

In that case, if X_i is one stdev above $\bar{x}$, then we might also guess that Y_i is one stdev above its mean $\bar{y}$

X_i one stdev above $\bar{x}$ → guess Y_i one stdev above $\bar{y}$

If X and Y are positively correlated but the correlation is weak, then knowing that the value of X is large tells us that maybe Y is also large, but we're less sure about it now, because X & Y aren't always behaving in the same way

weak, pos correlation: X & Y sometimes big @ same time, not always

In that case, if X_i is one stdev above $\bar{x}$, we would guess that y_i is also above $\bar{y}$, but not by a whole stdev, because X_i being large doesn't guarantee that y_i is also large.

X_i one stdev above $\bar{x}$ → guess y_i < 1 stdev above $\bar{y}$

Writing the estimated regression line slope as

$\hat{\beta}_1 = r_{xy} \frac{\sqrt{S_y^2}}{\sqrt{S_x^2}}$ formalizes that idea.

If x and y are perfectly correlated (i.e. correlation = 1), then when x increases by 1 standard deviation ($\sqrt{S_x^2}$) our prediction for y increases by 1 standard deviation ($\sqrt{S_y^2}$)

X, Y perfect correlation, $r_{xy} = 1$

⇓

If x increases by 1stdev ($\sqrt{S_x^2}$), predicted value for $\hat{y}$ increases by $\hat{\beta}_1 * \Delta x$

$= \hat{\beta}_1 * \sqrt{S_x^2}$

$= r_{xy} \cdot \frac{\sqrt{S_y^2}}{\sqrt{S_x^2}} \cdot \sqrt{S_x^2}$

$= 1$

$= \sqrt{S_y^2}$

What if the correlation is weak? If the correlation = 0.25, then when x increases by 1 stdev ($\sqrt{S_x^2}$), our prediction for y goes up by 0.25 stdev ($\sqrt{S_y^2}$).

if $r_{xy} = 0.25$, when x increases by 1stdev ($\sqrt{S_x^2}$), predicted value for $\hat{y}$ increases by $\hat{\beta}_1 * \underbrace{\sqrt{S_x^2}}_{\Delta x}$

$= r_{xy} \frac{\sqrt{S_y^2}}{\sqrt{S_x^2}} \cdot \sqrt{S_x^2}$

$= 0.25 \sqrt{S_y^2}$

next: look at more computational examples